

Gradient descent logistic regression

GradientDescent in Logistic Regression with formulas and one small numerical example

Step 1: Logistic Regression Equation

In **Logistic Regression**, the predicted value is a **probability** given by:

$$h_{\theta}(x) = \frac{1}{1 + e^{-z}}$$

where

$$z = \theta_0 + \theta_1 x$$

Here:

- $h_{\theta}(x) \rightarrow$ predicted probability (between 0 and 1)
 - $x \rightarrow$ input feature
 - $\theta_0, \theta_1 \rightarrow$ model parameters (weights)
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Step 2: Cost Function

The cost (error) function for one training example is:

$$J(\theta) = -[y \log(h_{\theta}(x)) + (1 - y) \log(1 - h_{\theta}(x))]$$

and for all training examples (n samples):

$$J(\theta) = \frac{1}{n} \sum_{i=1}^n -[y_i \log(h_{\theta}(x_i)) + (1 - y_i) \log(1 - h_{\theta}(x_i))]$$

Step 3: Gradient Descent Update Rule

We update parameters to minimize cost:

$$\theta_j = \theta_j - \alpha \frac{\partial J(\theta)}{\partial \theta_j}$$

where:

- α = learning rate
- $\frac{\partial J(\theta)}{\partial \theta_j}$ = gradient (slope)

For logistic regression:

$$\frac{\partial J(\theta)}{\partial \theta_j} = \frac{1}{n} \sum_{i=1}^n (h_{\theta}(x_i) - y_i) x_{ij}$$

So,

Gradient Descent Update Rule:

$$\theta_j := \theta_j - \alpha \frac{1}{n} \sum_{i=1}^n (h_{\theta}(x_i) - y_i) x_{ij}$$

Step 4: Simple Numerical Example

Let's take a tiny dataset:

x	y
1	0
2	1

Initial parameters:

$$\theta_0 = 0, \theta_1 = 0$$

Learning rate: $\alpha = 0.1$

Iteration 1

Step 1: Compute predictions

$$z_1 = 0 + 0(1) = 0 \Rightarrow h_{\theta}(x_1) = \frac{1}{1 + e^0} = 0.5$$

$$z_2 = 0 + 0(2) = 0 \Rightarrow h_{\theta}(x_2) = 0.5$$

Predictions: [0.5, 0.5]

Step 2: Compute gradients

For θ_0 :

$$\frac{\partial J}{\partial \theta_0} = \frac{1}{2}[(0.5 - 0) + (0.5 - 1)] = \frac{1}{2}(0.5 - 0.5) = 0$$

For θ_1 :

$$\begin{aligned}\frac{\partial J}{\partial \theta_1} &= \frac{1}{2}[(0.5 - 0)(1) + (0.5 - 1)(2)] \\ &= \frac{1}{2}(0.5 - 1) = \frac{1}{2}(-0.5) = -0.25\end{aligned}$$

Step 3: Update parameters

$$\theta_0 := 0 - 0.1(0) = 0$$

$$\theta_1 := 0 - 0.1(-0.25) = 0.025$$

✅ After 1 iteration:

$$\theta_0 = 0, \quad \theta_1 = 0.025$$

If you continue multiple iterations, θ_1 will keep adjusting until the cost is minimized and the predictions get closer to actual labels (0 or 1).



Summary

Step	Formula	Meaning
Hypothesis	$h_{\theta}(x) = \frac{1}{1+e^{-z}}$	Predict probability
Cost Function	$J(\theta) = -[y \log(h) + (1 - y) \log(1 - h)]$	Measure error
Gradient	$\frac{\partial J}{\partial \theta_j} = (h - y)x_j$	Direction of steepest ascent

Step	Formula	Meaning
Update Rule	$\theta_j := \theta_j - \alpha(h - y)x_j$	Move in direction of minimum
